PROJECT SCHOLARX

THE AI-ENABLED DOCTORATE

Research Smarter. Think Deeper. Create Impact.

PARTICIPANT WORKBOOK



AI can make you faster.
Only you can make your research meaningful.

Project ScholarX • Sprint 4 Companion Resources

akhil2453.github.io/akhil-portfolio

How to Use This Workbook

Use this workbook during the webinar to convert one real professional challenge into a preliminary doctoral research direction. It is intentionally designed for rough thinking, not final answers. Your strongest research question will emerge through iteration, evidence and supervisory discussion.

AKHIL TIP #1 Do not begin with a fashionable title. Begin with a recurring, consequential problem that deserves evidence.

- ☐ Keep one organisational problem in focus throughout the session.
- ☐ Write imperfect first answers. Refine them after each mission.
- ☐ Treat every AI output as a proposal requiring verification.
- ☐ Do not upload confidential, personal or commercially sensitive data.
- ☐ Record what AI assisted with and what you decided yourself.

MISSION 01

From Experience to Doctoral Inquiry

Turn professional observation into a researchable problem.

1. My Starting Organisational Problem

Complete the sentence in practical language. Avoid academic jargon at this stage.

A recurring problem I observe is:

This matters because:

The people or groups affected are:

The decision that better evidence could improve is:

2. Doctoral Thinking Ladder

Stage	My note
Observe	
Question	

Stage	My note
Investigate	
Evaluate	
Argue	
Create impact	

Observe: What pattern, failure, tension or unexplained outcome keeps recurring?

Question: What do I need to understand rather than merely assume?

Investigate: What literature, data and method could produce credible evidence?

Evaluate: What alternative explanations must be considered?

Argue: What claim can I defend from the evidence?

Create impact: What changes in understanding, decision or practice could follow?

3. PhD or DBA Alignment Check

Question	My current inclination
Is my primary contribution aimed at advancing theory or improving a complex professional situation?	
Do I have meaningful access to the organisational context, stakeholders or evidence?	
Will the outcome need to be academically defensible and practically implementable?	
Am I prepared to examine my own professional assumptions critically?	
Which audience should benefit most: a discipline, an organisation, a profession, or several of these?	

REMEMBER A DBA is not an easier PhD. It is a rigorous professional doctorate with a different purpose and form of contribution.

MISSION 02

The Business Research Galaxy

Scan functions, subdivisions and cross-functional tensions.

Function	Subdivisions	Illustrative problems
Finance	FP&A; corporate finance; treasury; risk; credit; audit; taxation; investment; fintech	Forecast bias; capital allocation; fraud; credit opacity; working-capital decisions; financial resilience
Marketing	Brand; digital; CRM; pricing; retail; consumer behaviour; customer experience; sales	Customer trust; churn; attribution; personalisation; channel conflict; loyalty; pricing fairness
Human Resources	Talent acquisition; performance; L&D; compensation; culture; leadership; wellbeing	Attrition; skills gaps; hybrid work; inclusion; leadership pipelines; burnout; algorithmic hiring
Operations	Process design; quality; maintenance; capacity; productivity; lean; service operations	Bottlenecks; quality variance; downtime; service delay; technology adoption; productivity paradox
Supply Chain	Procurement; inventory; logistics; demand planning; supplier relations; resilience	Forecast error; supplier risk; last-mile cost; bullwhip effect; visibility; disruption response
Strategy	Business models; transformation; governance; innovation; M&A; internationalisation	Execution gaps; digital transformation; ecosystem strategy; scaling; governance; strategic agility
Technology	AI; analytics; cybersecurity; cloud; IoT; automation; platforms; data governance	Trust; adoption; explainability; privacy; technical debt; data quality; cyber resilience
Education & Healthcare	Learning analytics; curriculum; assessment; faculty development; hospital operations; patient experience	Student success; faculty adoption; clinical workflow; patient safety; telehealth; service quality
Entrepreneurship & Sustainability	Start-up scaling; funding; innovation; ESG; circular economy; climate risk	Founder decisions; scaling failure; greenwashing; sustainable procurement; impact measurement

4. Research Opportunity Scan

Function / context	Recurring problem	Stakeholders	Decision affected

Function / context	Recurring problem	Stakeholders	Decision affected

5. Is the Problem DBA-Worthy?

☐ It is recurring rather than a one-time incident.

☐ There are competing explanations or unresolved tensions.

☐ The problem connects to scholarly literature and concepts.

☐ The scope can be bounded within a realistic doctoral study.

☐ It has significant organisational, professional or societal consequences.

☐ Credible evidence can be accessed ethically and feasibly.

☐ A better explanation could change an important decision or practice.

☐ I am willing to discover that my initial explanation may be wrong.

MISSION 03

Problem-to-Research Direction Canvas

Build a bounded, defensible preliminary direction.

Canvas field	Guiding question	My draft
Professional context	Industry, organisation type, geography, function or setting	
Observed problem	What repeatedly happens, for whom, and with what consequence?	
Current explanation	What do people currently believe causes the problem?	
Uncertainty or tension	What is not known, contested, contradictory or poorly understood?	
Stakeholders	Who experiences, influences or decides around the problem?	
Possible concepts / variables	What factors, mechanisms, behaviours or outcomes may matter?	
Evidence access	What documents, participants, systems or datasets may be available?	
Boundary	What will be excluded by context, time, population, function or geography?	
Practical significance	Which decision, policy, process or capability could improve?	
Scholarly significance	What understanding could be extended, tested or contextualised?	

6. Preliminary Research Question Builder

Use the structure below as scaffolding, not as a mandatory formula.

QUESTION FRAME How / why / to what extent does [phenomenon, mechanism or intervention] influence [outcome] among [population or stakeholder] in [defined context], and what are the implications for [decision or practice]?

Draft question 1:

Draft question 2:

7. Research Question Quality Check

Criterion	Diagnostic question	Rating 1–5
Clarity	Can a reader identify the central phenomenon, outcome and context?	
Researchability	Can evidence be collected or analysed to answer it?	
Scope	Is it narrow enough for a doctorate without becoming trivial?	
Feasibility	Are access, time, ethics and capability realistic?	
Practical relevance	Could the answer improve an important professional decision?	
Scholarly relevance	Does it connect to a meaningful body of literature or conceptual tension?	
Neutrality	Does it avoid assuming the conclusion before the study begins?	

MISSION 04

AI Research Command Centre

Use AI as an assistant inside a human-governed workflow.

8. The Human–AI Contract

AI may assist with	The scholar must retain
Brainstorming keywords, synonyms and alternative framings	Problem ownership and research purpose
Organising notes and suggesting structures	Theoretical judgement and interpretation
Explaining methods and generating practice code	Method selection, validation and reporting
Summarising provided text for orientation	Direct reading and source verification
Editing clarity, grammar and presentation	Originality, authorship and accountability
Generating charts or code from verified data	Data integrity, assumptions and analytical decisions

AKHIL TIP #2 Every AI-supported step should end with a human decision gate: verify, select, reject, revise or disclose.

9. Prompt Design Formula: R-C-T-C-O-V

Element	What to specify
Role	The perspective the AI should adopt, without pretending to be the author or decision-maker.
Context	Your industry, problem, population, constraints and current stage.
Task	The exact cognitive or production task required.
Constraints	Boundaries, exclusions, ethical limits and what must not be invented.
Output	Required structure: table, questions, checklist, comparison or critique.
Verification	What evidence, assumptions or uncertainties the model must flag for human checking.

10. Hands-on Activity: From Business Problem to Research Direction

Copy this prompt into an approved AI tool and replace the bracketed text.

01 Problem to research direction

Use when: You have a real problem but the direction remains broad.

> I am a professional working in [industry/function/context]. A recurring organisational challenge I observe is [problem]. Help me develop five possible DBA-level research questions. For each question, provide: (1) the central phenomenon or variables, (2) stakeholders, (3) practical significance, (4) a possible qualitative, quantitative or mixed-method approach, and (5) a clear scope boundary. Do not invent evidence, references or facts. Flag assumptions and ask up to three clarification questions before generating the final options.

HUMAN DECISION GATE: Select or reject options based on meaning, access, literature and feasibility.

AI output I want to retain or adapt:

What the AI misunderstood, assumed or made too generic:

My revised research question:

MISSION 05

Literature Exploration

Move from keywords to a traceable search strategy.

11. Concept and Keyword Planner

Search block	Core term	Synonyms / alternatives	Exclusions / boundaries
Phenomenon / problem			
Context / industry			
Population / stakeholder			
Mechanism / intervention			
Outcome / impact			

12. Boolean Search Builder

Combine synonyms with OR. Combine concept blocks with AND. Use quotation marks for phrases and database-specific field codes where appropriate.

PATTERN ("employee attrition" OR turnover OR retention) AND ("middle managers" OR "mid-career employees") AND (technology OR IT) AND (India OR "emerging markets")

My search string:

13. Literature Map

Theme / cluster	Seminal or central papers	Recent developments	Contradictions / gaps	Implication for my study

14. Source Verification Record

Claim or citation proposed by AI	Located in legitimate database?	Metadata verified?	Read relevant passage?	Decision

MISSION 06

Ethical AI Check

Protect confidentiality, originality and research integrity.

15. Traffic-Light Classification Activity

Scenario	Green / Amber / Red	Reason
Using AI to propose alternative keywords for a database search		
Asking AI to generate ten references and copying them without verification		
Uploading identifiable employee interview transcripts to a public AI service		
Using AI to improve grammar in a paragraph you wrote, subject to institutional policy		
Allowing AI to select a methodology without examining assumptions or feasibility		
Using AI to generate Python code, validating the code and reporting the analytical decisions		
Submitting AI-generated interpretation as your own without disclosure		
Using a source-grounded tool to compare papers, then checking the original papers		

16. Responsible AI Checklist

☐ I followed my university, supervisor, publisher and ethics-committee requirements.

☐ I did not upload confidential, identifiable or copyrighted material without authority.

☐ I verified every citation, quotation, statistic and factual claim in the original source.

☐ I can explain and defend every methodological and analytical decision.

☐ I did not list an AI system as an author or transfer accountability to it.

☐ I recorded the tool, date, purpose, prompt or workflow when disclosure may be required.

☐ I checked outputs for bias, omission, overconfidence and fabricated detail.

☐ The final reasoning, interpretation and wording represent my own accountable scholarly work.

17. AI Use Log

Date	Tool / version	Purpose	Input type	What I retained	Verification / disclosure

MISSION 07

From Scholar to Scholar–Practitioner

Turn the webinar into a next action.

18. Preliminary Research-Interest Statement

Use this structure for a counselling discussion, supervisor conversation or early concept note.

Professional context:

Recurring problem and why it matters:

Stakeholders and context:

What I want to understand or explain:

Evidence I may be able to access:

Possible contribution to practice and knowledge:

19. My 30-Day Research Upgrade Plan

Week	Action	Evidence of completion
Week 1	Clarify the problem and discuss it with a knowledgeable stakeholder.	
Week 2	Build and test a literature search strategy; save verified papers in a reference manager.	
Week 3	Create a literature map and revise the research question.	
Week 4	Discuss scope, methodology, access and ethics with a supervisor or research mentor.	

CONTINUE THE JOURNEY

Project ScholarX Resource Hub

Prompts, checklists, tools and future updates.

Your next step is not to ask AI to write more. It is to ask a better research question.

Visit the resource hub for the prompt library, toolkit and updates.

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SCAN TO CONTINUE